

National College of Ireland**Project Submission Sheet**

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Programme: MSc Cybersecurity **Year:** 2025-2026
Module: AI/ML in Cybersecurity
Lecturer: Taylou Maniganze
Submission Due Date: 20/02/2026
Project Title: CA1-proposal
Word Count: 1435

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard supplied by the Library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

Signature: Ahilya Sarnaik

Date: 20/02/2026

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**

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Signature:	Ahilya Sarnaik
Date:	20/02/2026
Penalty Applied (if applicable):	

AI Acknowledgement Supplement

AI/ML in Cybersecurity

Your Name/Student Number	Course	Date
Ahilya Sarnaik		20/02/2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

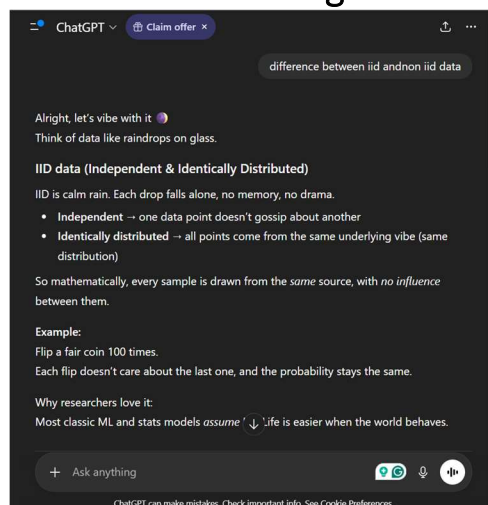
Tool Name	Brief Description	Link to tool
Chatgpt	Chat Generative Pre-Trained Transformer and was developed by an AI research	Chatgpt.com

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

Chatgpt	
Brainstorming	
difference between iid and non iid data	IID data (Independent & Identically Distributed) IID is calm rain. Each drop falls alone, no memory, no drama. Independent → one data point doesn't gossip about another

Evidence of AI Usage



Federated Learning with Adaptive Aggregations for anomaly detection comparing Intrusion Detection

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H9AIMLC - AI_ML in Cybersecurity

National College of Ireland

1. Introduction -Research Problem Background

As the amount of internet connectivity has been increased in few decades the surface for malicious threat actor has been increased making intrusion detection systems the most precious aspect in cybersecurity. Traditionally intrusion detection systems rely on a centralized architecture in which all traffic data is collected and analysed at an individual point. (Praveen Kumar Myakala, 2025). While this being effective with individual isolated deployment it comes with limitation if the single point fails, generates a substantial communication overhead and it surrenders sensitive network traffic to individual central authority and it becomes incompatible with modern data privacy regulations (Khraisat et al., 2024). The federated learning address the limitation by enabling the distributed clients to collaborate and train shared global model without sharing raw resources, only the model updates are shared with central servers.

While using federal learning the tapering FL algorithms like federal averaging (FedAvg), thinks that the client data is independently and identically distributed but in real network this assumption is violated cause different organization collects different traffics producing heterogeneous Non-IID distribution of data cross the clients (Karimireddy et al., 2020). Under this circumstances local model drifts towards client drift phenomenon causing naïve to produce a low graded global model, this can cause problems in practical anomaly detection for network intrusion, where the traffic is distributed and may exist in subnet of clients. (Bithi et al., 2026)

As a principal solution an adaptive aggregation strategy can be used by dynamically weighting each client's contribution based on distributional proximity or similarity in gradients to other global model rather than dataset size only. (Li et al., 2022) Performing comparative analysis with controlled IID and Non IID conditions it remains the motive of present study.

2. Summarized Lit Review/Justification

The body of knowledge surrounding Federated Learning (FL) for network security has become increasingly more developed from the year 2020 onwards. Most recently published the first systematic review of FL-based IDS, identifying data heterogeneity and aggregation strategies as the most important factors influencing performance of IDS and thus, directly motivating the emphasis on aggregation in this work. (Khraisat et al., 2024) Additionally, report on the FL-IDS implementations in the IoT, healthcare, and industrial domains, described the aggregation strategies, and reported that the presence of non-IID data remains the biggest obstacle in the deployment of FL-IDS. (Praveen Kumar Myakala, 2025)

The most thorough analysis for the non-IID problem was conducted by (Bithi et al., 2026), in which they benchmarked FedAvg, FedProx, SCAFFOLD and FedNova under 6 different strategies of heterogeneity partitioning and showed that there was severe loss in accuracy and convergence under conditions of high skew (Dirichlet $\alpha = 0.1$). This provides the benchmark for the comparisons of experimental results for the IID and Non-IID conditions of this study. (Karimireddy et al., 2021) SCAFFOLD is designed to address client drift. It uses control variates to reduce bias from the heterogeneous local gradients — which is one of the primary justifications for implementing adaptive aggregation in this work. It is in the field of anomaly

detection, where (Laridi et al., 2024) showed that federated thresholding for autoencoder reconstruction errors is considerably better than thresholding at the local level.

3. RESEARCH QUESTION

“To what extent an adaptive aggregation in federated learning improve anomaly detection performance for network intrusion detection under non-IID data conditions compared to IID conditions?”

There are three relevant factors that suggest this question is worth the investigating. Firstly, the understanding of FL-based IDS systems that work in the IID controlled environment labs, but are very bad in the real heterogeneous deployments, which becomes operational problem to the network defenders. Second, attempting to quantify the performance delta in the presence of IID, Non-IID, and showing that adaptive aggregation can reduce that delta is in a way, a contribution to the theory of FL convergence under distributional shift in the case of aggregation. In Proximal and Gradient Similarity Based Weighted Regularization there will be adjustable degrees of control that respond to the varying degrees of heterogeneity. The effects of heterogeneity of the parameters, controlled by regulation are to this point unknown concerning the phenomenon of anomaly detection. The last contribution to the field is that the researcher has developed a methodology that is replicable. The methodology includes everything from the established partitioning protocols, aggregated weight calculation, and the established evaluation metrics (AUROC, F1, convergence rounds) of the datasets, NSL-KDD and UNSW-NB15, that future FL-IDS researchers can build.

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